

Python-on-Windows Solution · Regulatory Compliance · Autoresearch Evolution · 100% On-Premise

Python-on-Windows · GoBD-compliant · 8-step pipeline

2 PREPROCESS

OpenCV 4.10 + Pillow
Deskew · binarize · denoise · enhance

Adaptive Sauvola binarization, Hough-transform deskew, CLAHE contrast enhancement. Parameters optimized by Autoresearch loop A1 on synthetic data. Handles faded thermal receipts and low-quality faxes.

pytesseract / EasyOCR

Tesseract for printed text (German + English default). EasyOCR for handwriting/degraded scans. Config auto-tuned by Autoresearch loop A2. Outputs HOCR with word-level bounding boxes + confidence scores.

Custom + OpenCV

Identifies header, address block, line items, totals, footer. Table extraction via line detection + cell segmentation. Autoresearch A4 tunes detection thresholds. Maps zones to extraction fields.

Ollama (local LLM)

Structured field extraction via optimized prompts

PyQt6 Desktop UI

§ 57 StBerG: MANDATORY. Side-by-side: original scan + OCR text + extracted fields. Low-confidence words highlighted in red. Amounts flagged if OCR confidence <80%. 'Freigegeben von' + timestamp logged.

ocrmypdf → PDF/A-2b

GoBD-compliant archival · immutable · hashed

Sandwich PDF (image + invisible text layer). PDF/A-2b for 10-year archival (\$147 AO). SHA-256 integrity hash per document. JBIG2 compression for B&W, JPEG for color. Compression tuned by A3.

DATEV DMS / Flatfile

Kanzlei system integration · CSV / XML

Outputs to DATEV import folder, Agenda, or flat CSV/XML. Extracted fields mapped to DATEV Buchungssatz format. Includes: Belegdatum, Belegnummer, Betrag, Konto, Gegenkonto, Steuerschlüssel.

All local · open source · zero cloud dependencies

pytesseract	OCR engine wrapper	Tesseract 5.x binary
easyocr	Deep learning OCR	80+ langs, GPU/CPU
ocrmypdf	OCR → PDF/A pipeline	Industry gold standard
opencv-python	Image processing	4.10+ headless
Pillow	Image primitives	PIL fork, 11.0+
scikit-image	Sauvola, morphology	Advanced algorithms
pilkeydf	PDF manipulation	QPDF-based, fast
PyMuPDF	PDF rendering/search	MuPDF engine
ollama	Local LLM inference	llama3, mistral, phi-3
NAPS2	Scanner frontend	eSCL/WIA/TWAIN CLI
PyQt6	Desktop review UI	Native Qt widgets
typer	CLI automation	Click-based, typed
jbig2enc	B&W PDF compress	JBIG2 encoding
deskew	Skew angle detect	Hough/projection
reportlab	PDF generation	Poster & reports

~~OpenAI API (ChatGPT, GPT-4, Whisper) - US cloud, § 203 violation~~

- ✗ Anthropic API (Claude) — US cloud, CLOUD Act exposure
- ✗ Google Cloud Vision / Document AI — US processor, Schrems II
- ✗ AWS Textract / Comprehend — US cloud, no § 203 agreement
- ✗ Azure AI Document Intelligence — US parent, CLOUD Act
- ✗ Any cloud-hosted OCR service — data leaves Kanzlei network
- ✗ External Streamlit / web hosting — exposes documents to Internet
- ✗ Hugging Face Inference API — sends data to external servers

PDF/A-2b	Archival format output (GoBD Rn.136)
300 DPI	Minimum scan resolution (BSI RESISCAN)
10 years	Retention: Belege, Rechnungen (§ 147 AO)
6 years	Retention: Geschäftsbriefe (§ 147 AO)
5 years	Retention: GoG KYC documents (GoG § 8)
SHA-256	Integrity hash per archived document
€0	Total software cost — 100% open source

Eingangsrechnungen	A5 Extract	A2 OCR	Betrag, USt, Datum accuracy
Lohnsteuerbeschein.	A4 Layout	A1 Preproc	Table/zone detection F1
Kontoauszüge	A2 OCR (numbers)	A4 Layout	Number CER specifically
Handschriftl. Belege	A1 Preprocess	A2 EasyOCR	Overall CER on handwriting
Finanzamtsbescheide	A4 Layout	A5 Extract	Structured form fields
Verträge	A5 LLM Summary	A2 OCR	Compress/Summary quality + size
Quittungen	A1 Preprocess	A2 OCR	CER on faded thermal paper

■ Schutzbedarfsanalyse per document type (TR-RESISCAN)

- AV-Verträge reviewed — no cloud processors confirmed
- Verarbeitungsverzeichnis updated (DSGVO Art. 30)
- Löschkonzept implemented (6/10-year auto-flagging)
- Berufshaftpflicht insurer notified of AI workflows
- Mandanteninformation issued (§ 62 StBerG)
- Staff trained: scan quality check + review duties
- BitLocker / FileVault verified on all workstations
- § 203 Verpflichtungserklärung signed by all staff

Every legal obligation mapped to a concrete technical rule and pipeline impact

A1

All autoresearch loops run **EXCLUSIVELY** on synthetic data (Faker invoices, ICDAR benchmarks). Never client documents.
Only optimized **CONFIG FILES** cross into production. Production pipeline uses human-in-the-loop review (§ 57 StBerG).
Research environment is fully isolated — no client data exposure, no § 203 applicability, no supervision required.

PHASE 1 Static Pipeline	CURRENT	PHASE 2 OCR Autoresearch	QUICK WIN	PHASE 3 Full Integration	TARGET	PHASE 4 Feedback Loop	VISION
Manual ocrmypdf defaults. OCR improvement loop. Baseline CER ~8-15%. POPIA output only.		A1 + A2 loops active. 15-50% CER improvement. CPU only, no GPU needed. ~500 experiments/night.		All A1-A6 loops active. LLM outputter ~90% acc. GPU recommended for A5. Full field extraction.		Anonymized prod metrics. Self-improving pipeline. Adapts to new doc types. Drift detection.	